STAGE 8–10 · CONSOLIDATED

Jura MOGP Findings

Multi-output Gaussian process prediction of cadmium in the Swiss Jura, after the exact-Cholesky correction. Every number below supersedes the corresponding figure in notebooks 03, 04 and the stage-7 sweep.

Data Jura, 359 locations, Cd/Ni/Zn **Design** heterotopic, 259/100

Selected model LMC, Q = 2

RMSE IMPROVEMENT

+27.3%

LMC over independent GP, mean of 10 random splits (±4.5%)

SPLITS WON

10 / 10

Paired t p = 6×10^{-8}. The MOGP advantage is not luck.

LMC OVER ICM

5 / 10

p = 0.19, not significant. The gain is interpretive, not predictive.

What the correction changed

Why every earlier number moved, and by how much

GPyTorch computes the exact marginal likelihood by Cholesky factorisation only while the training set stays under `max_cholesky_size`, which defaults to 800. This project trains on 977 observations, so every multi-output model up to stage 7 was silently using conjugate gradients with a stochastic Lanczos log-determinant estimate.

Evaluating the same model at the same parameters five times measures the damage directly:

COMPUTATION	REPEATED EVALUATIONS OF −LOG ML / N					SPREAD
CG + Lanczos WAS USED	1.490759	1.488572	1.482091	1.481232	1.481291	0.0095

Exact Cholesky NOW	1.485629	1.485629	1.485629	0.0

Noise band 0.0095. The stage-7 sweep selected Q from loss differences of 0.0006 — a signal-to-noise ratio below 1 : 15.

NOT JUST NOISE — BIAS

Re-running the Q sweep under exact arithmetic raised every loss by a consistent **+0.023 to +0.030**. The Lanczos estimator systematically overstated the log-likelihood, so the old values cannot be defended as unbiased-but-noisy.

Monotonicity restored

Training marginal likelihood must improve as parameters are added. Under CG it did not — Q=3 scored worse than Q=2, and Q=5 worse than Q=4. Under exact arithmetic the sequence is monotone, which is the check that confirms the fix rather than a result in itself.

Q	LOSS (CG, STAGE 7)	LOSS (EXACT)	SHIFT	ORDERING
1	~~0.9815~~	1.004174	+0.0227	—
2	~~0.9182~~	0.948543	+0.0303	↓
3	~~0.9188~~ ROSE	0.947921	+0.0291	↓
4	~~0.9127~~	0.942613	+0.0299	↓
5	~~0.9132~~ ROSE	0.942559	+0.0293	↓

Choosing Q

Training likelihood cannot select a model; BIC and AIC agree it is Q = 2

Because the exact loss is now monotone in Q, it selects Q_{max} by construction and is useless as a criterion. Test RMSE would leak the test set. Both information criteria penalise the parameter count `k = 8Q + 2` against N = 977 and land on the same answer.

Q	K	−2 LOG ML	BIC	AIC	TEST RMSE
1	10	1962.2	2031.0	1982.2	0.7428

2	18	1853.5	1977.4	1889.5	0.7313
3	26	1852.2	2031.2	1904.2	0.7323
4	34	1841.9	2075.9	1909.9	0.7198
5	42	1841.8	2130.9	1925.8	0.7172

BIC at Q=2 leads the runner-up by 53 units. Corroboration: at Q=5 the fifth latent function is assigned 0.0% of the variance in all three outputs — the optimiser switches it off unprompted.

> A NOTE ON THE EARLIER Q = 2
>
> Stage 7 also chose Q = 2, but from test-RMSE differences that were smaller than the CG noise band. The answer survives; the reasoning does not. Report the BIC/AIC argument, not the original one.

Two spatial scales

The central substantive result, and the reason ICM is not sufficient

The selected model splits the covariance into a short-range and a long-range latent function with sharply different cross-output correlations.

LATENT	ℓ_X (KM)	ℓ_Y (KM)	CD VAR	NI VAR	ZN VAR	CD–NI CORR
q = 1 short	0.052	0.042	77.7%	46.5%	68.8%	0.673
q = 2 long	0.511	0.347	22.3%	53.5%	31.2%	0.941

Noise 0.0847, constant mean 0.0506. Decomposition of the posterior mean verified additive to 1.3×10^{-12}.

The long-range component carries near-perfect cross-correlations of 0.85–0.94, meaning the three metals behave as scalings of one shared field at that scale — the signature of a common regional source. The short-range component correlates only around 0.67 and is metal-specific, and it dominates cadmium's variance.

Note that the variance shares invert once the data speak: measured on the prediction surface itself, the long-range function supplies **62.9%** of the posterior mean's amplitude against the short-range function's 37.1%, despite holding only 22.3% of cadmium's prior variance.

What ICM loses

MODEL	LENGTHSCALE (KM)	CD–NI CORRELATION
ICM — its only one	(0.079, 0.058)	0.743
LMC q = 1	(0.052, 0.042)	0.673
LMC q = 2	(0.511, 0.347)	0.941

THE ARGUMENT FOR LMC

Constrained to a single lengthscale, ICM lands on (0.079, 0.058) — essentially the short-range component — and the long-range structure disappears entirely. Its single correlation of 0.743 is a variance-weighted blend of two physically distinct couplings, 0.673 and 0.941, and corresponds to no real process. That, not the 1.5% RMSE margin, is what justifies the full LMC.

THIS OVERTURNS AN EARLIER CONCLUSION

Notebooks 03 and 04 read the collapsed lengthscale as evidence that cadmium has weak spatial autocorrelation in Jura. It does not. Long-range structure was present the whole time; ICM's single-scale constraint suppressed it, and the optimisation was dominated by the 718 fully-observed Ni/Zn records.

Borrowing strength is strictly local

Where the multi-output advantage actually comes from

Posterior uncertainty was evaluated on a 140×140 grid and binned by distance to the nearest sampling location — which, because Ni and Zn are measured at all 359 sites, is the distance to the nearest borrowable observation.

DISTANCE (KM)	GRID CELLS	GP STD	LMC STD	REDUCTION
0 – 0.02	290	0.6877	0.4921	+28.4%
0.02 – 0.05	1274	0.7098	0.6815	+4.0%

0.05 – 0.10	3929	0.7448	0.8587	−15.3%
0.10 – 0.20	5424	0.7880	0.9024	−14.5%
0.20 – 0.35	1996	0.9203	0.9326	−1.3%
0.35 – 0.60	2230	0.9451	0.9548	−1.0%
0.60 +	4457	0.9473	0.9754	−3.0%

Across the 100 held-out cadmium sites the reduction is 22.9% (GP 0.7521 ± 0.1226 → LMC 0.5799 ± 0.1206).

The crossover falls between 0.05 and 0.10 km — precisely the short-range lengthscale of 0.052 km. The borrowing radius *is* $\ell_{q=1}$. Beyond it the LMC is the less confident of the two models, and it is right to be: the independent GP buys uniform mid-level confidence everywhere by inflating both its lengthscale and its noise. The LMC's sharper honesty is what produces its much better NLL (1.218 against 1.552).

This matters for how the result is framed. The advantage does not come from spatial interpolation. It comes from the heterotopic design placing a co-located Ni and Zn measurement at every site where cadmium is missing.

Robustness across 10 splits

One result strengthens; one does not survive

MODEL	RMSE	MAE	NLL	GAIN OV
Independent GP	0.8694 ± 0.0620	0.6693 ± 0.0412	1.2761 ± 0.1310	
ICM	0.6399 ± 0.0619	0.4802 ± 0.0470	0.9873 ± 0.1570	+26.3% ±
LMC (Q = 2)	0.6320 ± 0.0544	0.4696 ± 0.0424	0.9662 ± 0.1360	+27.3% ±

Each split re-randomises the 259/100 partition and recomputes the standardisation from that split's training observations only. Q is held at 2 rather than re-selected per split.

PAIRED COMPARISON	MEAN Δ RMSE	PAIRED T	WILCOXON	WINS	VERDICT
GP vs LMC	+0.2374	6.0×10^{-8}	0.0020	10/10	ROBUST
GP vs ICM	+0.2295	2.6×10^{-7}	0.0020	10/10	ROBUST

ICM vs LMC	+0.0079	0.186	0.375	5/10	NOT SIGNIFICANT

RETRACT THIS CLAIM

The single-split figure of "LMC beats ICM by 1.55%" does not replicate. Across ten splits the LMC wins five and loses five, at p = 0.19. The report must not claim that the full LMC predicts cadmium better than ICM. Its case rests entirely on the two-scale decomposition being recoverable and interpretable — which ICM structurally cannot represent.

Numbers that must not be reused

Every superseded value, with its replacement

SOURCE	CLAIM	STATUS	REPLACEMENT
nb 03	ICM test RMSE ~~0.7418~~	CG	0.7428
nb 03	ICM lengthscale ~~(0.074, 0.058)~~	CG	(0.0785, 0.0578)
nb 03	B correlations ~~0.753 / 0.718 / 0.762~~	CG	0.743 / 0.726 / 0.751
nb 03	ICM noise ~~0.0912~~	CG	0.0937
nb 03	improvement ~~+23.06%~~	CG	+23.53% single split; +26.3% ± 5.4% over 10
nb 04	rank sensitivity table for ICM	MEASURES NOTHING	rank has *no* effect; see below
nb 04	"loss spread 0.0032 shows stable convergence"	MISREAD	that spread was CG noise; exact spread is 1.3×10^{-9}
nb 03/04	"mean sampling spacing 0.27 km"	WRONG STATISTIC	mean NN 0.125 km; grid 0.25 km; clusters ~0.01 km
nb 03/04	"lengthscale collapse ⇒ weak spatial structure"	OVERTURNED	long-range component exists; ICM masked it

stage 7	Q sweep losses	CG	all shifted by +0.03; see table above
stage 7	"Q = 2 best" (by RMSE)	RIGHT, BADLY ARGUED	keep Q = 2, argue from BIC and AIC
script 02	"log transform makes concentrations symmetric"	FALSE FOR NI	see skew table below

Rank in `IndexKernel` is not the LMC's Q

With $B = W W^{T} + \mathrm{diag}(\kappa)$ and P = 3 outputs, rank 1 already provides 3 + 3 = 6 free parameters against the 6 degrees of freedom of a 3×3 PSD matrix. Rank 1 therefore spans the whole feasible set and higher ranks add pure redundancy. Measured under exact arithmetic:

```
ICM rank = 1   loss 1.004174
ICM rank = 2   loss 1.004174
ICM rank = 3   loss 1.004174
LMC Q    = 1   loss 1.004174   spread 1.34×10⁻⁹
```

The stage-4 rank sensitivity analysis was measuring a quantity whose true value is identically zero. Replace that table with this identity argument — it is the stronger result.

The log transform helps Cd and Zn, and harms Ni

METAL	RAW SKEW	LOG SKEW	EFFECT
Cd	+1.478	−0.249	CORRECTED
Zn	+1.493	−0.061	CORRECTED
Ni	+0.111	−1.131	MADE WORSE

Nickel was already symmetric. Justify the transform on the grounds actually available — multiplicative geochemical processes, a common scale for the cross-covariance, and consistency across outputs — not on skew correction for all three.

The sampling design

Why two lengthscales are identifiable at all, and what that costs

The 0.27 km figure quoted earlier was √(A/N) over the bounding box — a density estimate, not a measurement. Jura's nearest-neighbour distances are strongly bimodal, so no single spacing exists.

STATISTIC	VALUE	READING
√(A/N)	0.2525 km	lands on the grid mode; source of the 0.27 error
mean NN distance	0.1251 km	falls in the empty valley between the two modes
NN p25	0.0143 km	nested infill clusters
NN p75 – p100	0.2496 – 0.2505 km	the regular survey grid
sites with a neighbour < 50 m	151 / 359	42% of the survey sits in clusters

The design has two characteristic scales, roughly 10 m within clusters and 250 m on the grid, and the model recovered two lengthscales of 47 m and 429 m. That correspondence is worth stating: the nested infill is what makes the short-range component estimable at all. A pure 250 m grid could not resolve a 50 m lengthscale.

> STATE THIS AS A LIMITATION
>
> Only **193 of 64,261** point pairs (0.30%) lie below the short-range lengthscale, and nearly all of them come from the infill clusters. The q = 1 estimate therefore rests on a small and spatially unrepresentative subset of the data.

What to claim

Defensible positions, and the ones to avoid

✓ Multi-output GPs substantially outperform an independent GP for cadmium on Jura: **+27.3% ± 4.5%** RMSE, winning all ten random splits at $p = 6\times10^{-8}$.

Report the ten-split interval as the headline, not the single-split value.

✓ The coregionalisation matrix recovers the empirical cross-correlations without being shown them — learned 0.729 / 0.717 / 0.744 against measured 0.654 / 0.686 / 0.699.

The systematic upward offset is expected: B describes noise-free latent correlation, while the Pearson values are diluted by observation noise.

✓ An LMC with Q = 2 resolves a long-range common factor (ℓ ≈ 0.43 km, cross-correlations 0.85–0.94) from a short-range metal-specific one (ℓ ≈ 0.05 km, 0.67), consistent with a regional geological background overlaid by local enrichment.

Both BIC and AIC select Q = 2 independently.

✓ The advantage is driven by co-location, not by spatial smoothing: the uncertainty reduction vanishes beyond 0.05 km from a sampling site, exactly the short-range lengthscale.

✗ Do not claim the full LMC predicts better than ICM.

5/10 splits, p = 0.19. Argue instead that ICM's single lengthscale conflates two distinct spatial processes and reports a correlation of 0.743 that belongs to neither.

✗ Do not claim cadmium has weak spatial autocorrelation in Jura.

That conclusion came from ICM's constrained fit. The LMC finds real long-range structure carrying 62.9% of the prediction surface's amplitude.

✗ Do not present the stage-4 rank sensitivity table.

Rank has no effect on ICM when P = 3, provably. Present the identity argument instead.

Also worth writing up

- The Hadamard long-format construction admits missing observations but forfeits the Kronecker speed-up — the two are mutually exclusive, which is precisely why realistic problems need SVGP.
- No `ScaleKernel` on the multi-output models: its outputscale and B's diagonal are jointly unidentifiable. The independent GP keeps one because it has no B.
- Latent lengthscale initialisations must be spread apart. Identical initialisations leave the gradients symmetric and every latent function converges to the same solution, silently degrading the LMC back to ICM.
- Model selection ran on the training marginal likelihood throughout; test data touched the pipeline only at final evaluation.

Provenance

Everything regenerable from these scripts; all training deterministic

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`src/05_true_lmc.py` exact-Cholesky sweep — GP, ICM rank 1–3, LMC Q 1–5, BIC selection
`src/06_spatial_maps.py` posterior maps, latent decomposition, uncertainty vs distance
`src/07_robustness.py` 10 splits, paired tests (`--retrain` to force)
`src/08_sampling_design.py` nearest-neighbour and lag analysis